

1. DATA ENCRYPTION STANDARD (DES)

AIM

To implement the Data Encryption Standard (DES) algorithm using Python.

Algorithm

1. Start the program.
2. Import the required DES library.
3. Enter the plaintext.
4. Enter an 8-character secret key.
5. Pad the plaintext to make it a multiple of 8 bytes.
6. Encrypt the plaintext using DES.
7. Display the encrypted ciphertext.
8. Decrypt the ciphertext.
9. Display the original plaintext.
10. Stop.

Python Code

```
from Crypto.Cipher import DES
from Crypto.Util.Padding import pad, unpad
import base64

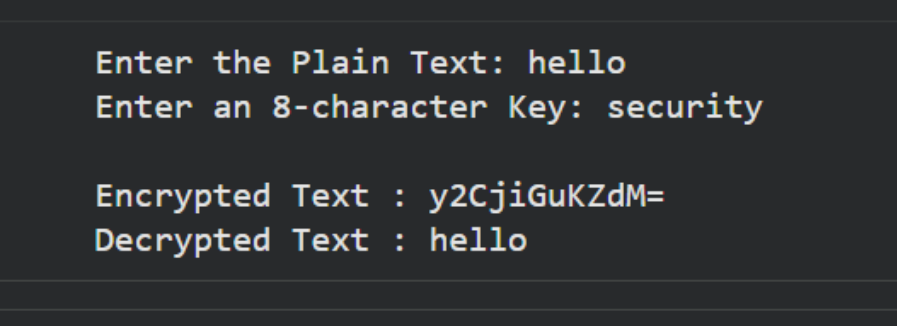
plaintext = input("Enter the Plain Text: ")
key = input("Enter an 8-character Key: ").encode()

cipher = DES.new(key, DES.MODE_ECB)

encrypted = cipher.encrypt(pad(plaintext.encode(), DES.block_size))
print("\nEncrypted Text :", base64.b64encode(encrypted).decode())
```

```
decrypted = unpad(cipher.decrypt(encrypted), DES.block_size)
print("Decrypted Text :", decrypted.decode())
```

Output



```
Enter the Plain Text: hello
Enter an 8-character Key: security

Encrypted Text : y2CjiGuKZdM=
Decrypted Text : hello
```

Result:

The above code is executed and the output is obtained successfully.